

TOP 20 EXAM TRAPS

- X **model="gpt-4o"** → ✓ model=DEPLOYMENT_NAME
- X **Private endpoint alone** → ✓ + disable public network access
- X **401 = missing role** → ✓ 401=bad auth · 403=missing RBAC role
- X **Container = offline** → ✓ Only DISCONNECTED container is offline
- X **Hallucination → Content Safety** → ✓ Hallucination → Groundedness detection
- X **Vector-only search for RAG** → ✓ Hybrid + semantic ranker
- X **Fine-tuning for facts** → ✓ Facts→RAG · Style/format→Fine-tuning
- X **Semantic Kernel is current** → ✓ MAF 1.0 is the current path
- X **Agent for regulated process** → ✓ Deterministic → Workflow orchestration
- X **requires_action = error** → ✓ Normal pause for custom fn output
- X **Same layout → neural model** → ✓ Same layout→template · Varies→neural
- X **Any CV domain exportable** → ✓ Only COMPACT domains export to ONNX
- X **Face API has emotion/age** → ✓ Retired for Responsible AI
- X **Identify = claimed identity** → ✓ Claimed → Verify(1:1) · Unknown→Identify(1:N)
- X **mixed sentiment on sentences** → ✓ "mixed" = document level ONLY
- X **recognize_once() for long audio** → ✓ start_continuous_recognition()
- X **Add filterable to live field** → ✓ Must rebuild + re-index
- X **dall-e-3** → ✓ Retired 4 Mar 2026 → gpt-image-2
- X **image.url from gpt-image** → ✓ gpt-image returns b64_json NOT url
- X **Video response is synchronous** → ✓ Sora 2 = async job → poll → download
- X **temperature + top_p together** → ✓ Use ONE only
- X **Transparent bg → JPEG/WebP** → ✓ PNG only (WebP unsupported on Azure)
- X **in_scope:false fixes over-refusal** → ✓ Lower strictness value instead
- X **Retry logic fixes sustained 429s** → ✓ Sustained 429 → Provisioned Throughput
- X **Knowledge store = search index** → ✓ Knowledge store → Power BI / analysis

RESPONSIBLE AI PRINCIPLES

Fairness	Reliability & Safety	Privacy & Security
Inclusiveness	Transparency	Accountability
Mnemonic: FRPITA · Content Safety harm categories: Hate · Violence · Sexual · SelfHarm · Severity: 0, 2, 4, 6		

SERVICE DECISION MAP

Scenario signal	Use
Generate text, reason, chat	Azure OpenAI (GPT models)
Text → embedding vectors	text-embedding-3-small/large
Agent with tools, persistent threads	Foundry Agent Service
Code-based agent orchestration	MAF 1.0 (agent-framework)
Search docs (keyword+vector)	Azure AI Search
Describe / tag / OCR an image	Vision (ImageAnalysisClient)
"What's wrong in this photo?" (reasoning)	Multimodal LLM (GPT-4o)
Generate image from text	gpt-image-2
Edit region, keep rest intact	Inpainting + mask
Generate video	Sora 2 (async job)
Extract invoice/receipt/ID fields	DI prebuilt model
Same-layout custom form	DI template model
Variable-layout custom form	DI neural model
Multiple doc types, one endpoint	DI composed model
Audio + video + image + doc together	Content Understanding
Text → extract schema (bespoke)	LLM + json_schema
NER / PII / Sentiment at scale	Azure AI Language
Tone / urgency / escalation	LLM classification
How it was SAID (frustration)	Audio-native multimodal
User intent in utterance	CLU
Answer FAQ questions	Question Answering
Route commands + FAQs	Orchestration Workflow
Distinguish diagnosed vs ruled-out	Text Analytics for Health (assertion)
Screen-reader description	Alt text via multimodal
Harmful image / text	Content Safety
Instructions hidden in image	Prompt Shields (indirect)
Scanned PDFs → searchable	Indexer + OCR skill
Chunk + embed at index time	Integrated vectorization
Custom logic during indexing	Web API (custom) skill

Enriched data → Power BI	Knowledge store	
Mark AI output as AI-generated	C2PA content credentials	
Consistent HV, no throttling	Provisioned Throughput (PTU)	
Cheap batch, results next day	Batch API	
No internet at all	Disconnected container	
DEPLOYMENT TYPES		
Type	Best for	
Standard	Default. Variable / low volume. Pay-per-token.	
Global Standard	Higher throughput. No data-residency req.	
Data Zone	Throughput + stay within EU/US boundary	
Provisioned (PTU)	Sustained high volume. No throttling. Fixed hourly.	
Batch	~50% cheaper. 24-hr async. Latency irrelevant.	
EVALUATION METRICS		
Metric	Tests	Scale
Groundedness	Answer supported by context?	1-5
Relevance	Answer addresses the question?	1-5
Retrieval	Did search return right chunks?	1-5
Coherence	Logically structured?	1-5
Fluency	Natural language?	1-5
F1 / BLEU	Match to reference answer	0-1
Violence/Hate	Harm categories	0-7
Key: Groundedness = hallucination · Relevance = off-topic · Retrieval = bad search · High retrieval + low groundedness = model ignoring good context		
AGENT SERVICE OBJECTS + RUN LIFECYCLE		
Object	Is	
Agent	Config: model, instructions, tools. Reusable.	
Thread	Conversation. Persists across turns.	
Message	One user/assistant turn in a thread.	
Run	One execution of agent against thread.	
Run states: queued → in_progress → completed requires_action failed cancelled expired		
requires_action = waiting for YOUR custom fn output (NOT an error)		
ORCHESTRATION PATTERNS		
Pattern	Signal words	
Sequential	"pipeline", "each step feeds next"	
Concurrent	"parallel", "same time", "reduce latency"	
Handoff	"route to specialist", "transfer to"	
Group Chat	"collaborate", "critique each other"	
Magentic	"manager plans + adapts", "open-ended goal"	
NOT an agent pattern Workflow	"identical steps every time", "auditable"	
RBAC ROLES (LEAST PRIVILEGE)		
Role	Can do	
OpenAI User	Call inference. Cannot deploy.	
OpenAI Contributor	Call + deploy models.	
AI Developer	Full dev in project. Cannot create hubs.	
AI Administrator	Hub management + networking.	
Search Index Data Reader	Query index only.	
Network: Private endpoint + disable public = fully blocked. 401=no auth. 403=no role.		
MODEL SELECTION		
Scenario	Model	
Complex reasoning, code	GPT-4o / GPT-5	
Multi-step logic, maths	o-series (reasoning)	
Cost-critical, simple tasks	GPT-4o-mini / Phi	
Edge / offline / no cloud	Phi (small model)	
Vectors for search	text-embedding-3-small	
Generate images	gpt-image-2 (current GA)	
Agent image tool	gpt-image-1 (required)	
Live voice	GPT-4o Realtime / Voice Live	
EXAM DAY		
✔ 700/1000 to pass (scaled score) · 120 min · ~2 min/question		
✔ Always ANSWER before flagging — zero penalty for wrong		
✔ Read LAST sentence of Q first (actual ask) → find the CONSTRAINT		

- ✓ Read each sentence or q first (usually) & find the constraint.
- ✓ Microsoft Learn open in-exam: max 2 lookups, costs 3-5 min each
- ✓ Flag & move on after 3 min on any single question
- ✓ Reserve last 15 min for flagged questions